P1DF7-00、P1DF7-F0、P1DF7-F1、P1DF7-F2、P1DF8-F0、P1DF8-F1、P1DF8-F2、P1DF9-F0、P1DF9-F1、 P1DF9-F2

Fault diagnosis code
P1DF7-00：Over Current Peak Error
P1DF7-F0：Discharge over current level 1
P1DF7-F1：Discharge over current level 2
P1DF7-F2：Discharge over current level 3
P1DF8-F0：Charge over current level 1
P1DF8-F1：Charge over current level 2
P1DF8-F2：Charge over current level 3
P1DF9-F0：Recharge over current level 1
P1DF9-F1：Recharge over current level 2
P1DF9-F2：Recharge over current level 3
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• BMS fault • BMS insulation fault • BMS high voltage circuit fault • The current circuit of BMS fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1DF7-00
The fuse may break due to excessive passing current:|bus circuit current| ⩾ 1000A
P1DF7-F0
Any branch discharge current is higher than the alarm threshold: ⩾ 1.05*look-up discharge current+1A
P1DF7-F1
Any branch discharge current is higher than the alarm threshold: ⩾ 1.2*look-up discharge current+1A
P1DF7-F2
Any branch discharge current is higher than the alarm threshold: ⩾ 1.25*look-up discharge current+1A
P1DF8-F0
Any branch charge current is higher than the alarm threshold: ⩽ 1.05*look-up charge current+1A
P1DF8-F1
Any branch charge current is higher than the alarm threshold: ⩽ 1.1*look-up charge current+1A

P1DF8-F2
Any branch charge current is higher than the alarm threshold: ⩽ 1.2*look-up charge current+1A
P1DF9-F0
The recharging flow of any branch is higher than the alarm threshold in driving mode: ⩽ 1.05*look-up recharge current+1A
P1DF9-F1
The recharging flow of any branch is higher than the protection threshold in driving mode: ⩽ 1.2*look-up recharge current+1A
P1DF9-F2
The recharging flow of any branch is higher than the safe threshold in driving mode: ⩽ 1.25*look-up recharge current+1A
To set the effect of a fault code condition

Troubleshooting steps	
Step 1	General inspection.
	1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 If there is any fault code except for P1DF7-00/P1DF7-F0/ P1DF7-F1/P1DF8-F0/P1DF8-F1/P1DF8-F2?
	YES → Refer to:DTC Summary list(BMS).
NO ↓	
Step 3	Check the connecting cable of BMS.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 4	Check the insulation of BMS.
	1 Use a megohm meter to check the insulation resistance of the BMS housing and grounding according to the procedure in the maintenance manual. **Standard value（insulation level 1）＞ 750Ω/V （DC charging ＞ 500Ω/V）** **Standard value（insulation level 2）＞ 500Ω/V （DC charging ＞ 100Ω/V）** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the insulation of BMS high-voltage cables.
	1 Use a megohm meter to check the insulation resistance of the BMS high voltage and grounding according to the procedure in the maintenance manual. **Standard value（insulation level 1）＞ 750Ω/V （DC charging ＞ 500Ω/V）** **Standard value（insulation level 2）＞ 500Ω/V （DC charging ＞ 100Ω/V）** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Perform initialization.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
	NO → Repair or change the BMS module.
YES ↓	

Step 7	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1000_1 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	